Noah Zhang

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Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Physics (Advisor: Surabhi Sachdev) — *Aug 2023 – Present (Exp. 2028)*

Georgia Institute of Technology — Atlanta, GA
M.S. in Computational Science and Engineering (CSE) — *Aug 2023 – May 2025*

Nanyang Technological University (NTU) — Singapore
B.Sc. in Physics (First Class Honors) — *Aug 2019 – May 2023*

- **GPA: 4.62/5.00** (Major GPA: 5.00/5.00). Awarded Full Ministry of Education Scholarship.

Technical Skills & Honors

Languages/Tools: C++ (STL, Boost), Python (NumPy, SciPy, Pandas, statsmodels), SQL; HPC, MPI, HTCondor, Git, Linux
Quantitative: MCMC, Bayesian Inference, MLE, Time-Series Analysis, Signal Processing, Alpha Research (WorldQuant BRAIN)
Statistical Learning: Ridge, Walk-Forward/Purged & Nested CV, PCA/SVD, k-Means, Bootstrap, Permutation Tests, FDR
Data: CRSP (WRDS), Ken French Data Library, high-frequency tick data
Honors: PLANCKS Physics Competition (Singapore) Silver Medal, 2022 | Odyssey Research Symposium Best Presentation, 2020

Research Experience

LIGO Scientific Collaboration (Georgia Tech) — Aug 2023 – Present
Graduate Research Assistant – Gravitational-Wave Detection & Inference — *Atlanta, GA*

- **Targeted Follow-up Search:** Co-developed a targeted search for the GstLAL pipeline that folds external electromagnetic triggers (arrival time, sky location) into the detection ranking statistic as **Bayesian priors**, reducing the false-alarm rate of associated candidates by **orders of magnitude** at unchanged latency.
- **Deployment:** Used in the **O4a** observing run and being deployed for the next run (IR1); now extending it to multi-pixel sky-localization priors. Second author of the method paper (under review, ApJ).
- **Real-Time Parameter Estimation:** Improved minutes-latency source parameter estimation from GstLAL low-latency SNR data by combining **MCMC**, **nested sampling** and prior updating, raising P-P plot calibration consistency in validation by **33%**.
- **IMBH Search (Lead, O4b/O4c):** Lead the offline intermediate-mass black hole search; improved the background noise model using **time-slid triggers**, making the likelihood-ratio ranking statistic more reliable. Contributed to the **GWTC-5** and **GWTC-6** catalogs.
- **Pipeline Engineering:** Maintain the GstLAL hybrid **C++/Python** pipeline, processing terabytes of streaming data with sub-second latency for real-time trigger generation.

Nanyang Technological University — May 2020 – Aug 2022
Undergraduate Researcher – Stochastic Modeling & Simulation — *Singapore*

- Modeled foraging dynamics with correlated random walks and time-delay ODEs; combined MCMC with a customized random-walk sampler, cutting simulation runtime from **days to hours (∼50x)**.

Publications & Presentations

L. Tsukada, **N. Zhang**, et al., *Fast targeted gravitational-wave followup search for compact binary mergers using GstLAL pipeline*, arXiv:2606.22266 (under review, ApJ).
N. Zhang and Yong, *Dynamics, Statistics, and Task Allocation of Foraging Ants*, Phys. Rev. E **108**, 054306 (2023).
LIGO–Virgo–KAGRA Collaboration, *GWTC-5* and *GWTC-6* catalog papers (contributor, IMBH search).
Talk: *Toward fast targeted gravitational-wave followup search for compact binary coalescences using GstLAL inspiral pipeline*, APS Meeting, Mar 18, 2026.

Selected Quantitative Projects

Cross-Sectional Equity Momentum Pipeline | *Python, CRSP/WRDS* | GitHub — Apr 2026 – May 2026

- Reproduced 12–1 momentum under a **pre-registered** plan (point-in-time S&P 500 universe rebuilt on CRSP, delisting returns, costs, single-shot 2020–2024 holdout); long-short returns correlate **0.90** in-sample and **0.83** out-of-sample with the Ken French UMD factor.

Walk-Forward Return Prediction with an Empirical Null | *Python, Ridge Regression* — Sep 2026

- Purged walk-forward ridge model for next-day SPY returns; out-of-sample IC **0.088**, **3.2**σ above a block-permutation null ($p < 0.005$).

WorldQuant Int'l Quant Championship 2026 – Stage 2, USA Region | *BRAIN* — Mar 2026 – May 2026

- Advanced past Stage 1 among **156,000+** registrants with **11** cross-sectional alphas (best four out-of-sample **Sharpe 1.3–2.0**); an ablation exposing a composite's Sharpe drop **1.59** $\rightarrow$ $-$**0.35** led to a standalone-Sharpe check before combining signals.